

EpiClock v4.0

DNA Methylation Epigenetic Age Analysis Platform

Publication-Ready Standards Implementation

Technical Documentation - 26 January 2026

ABSTRACT

EpiClock v4.0 is an advanced computational platform designed for DNA methylation-based epigenetic age analysis in addiction research. This document details the implementation of 11 major scientific publication standards, ensuring full compliance with international reporting guidelines for EWAS, biomarker validation, diagnostic accuracy, and data sharing.

NEW STANDARDS - 26 JANUARY 2026

1. EWAS Reporting Standards

- Epigenome-wide significance threshold: $P < 1 \times 10^{-7}$
- Bonferroni correction for EPIC v1: 5.77×10^{-8}
- DMR (Differentially Methylated Regions) analysis with Fisher's combined p-value
- methQTL integration and sensitivity analysis
- Cell-type deconvolution: Houseman, EpiDISH, ReFACTor methods
- Batch correction tracking: ComBat, SVA, RUVSeq
- 36-item comprehensive reporting checklist

2. MIQE Standards

- Minimum Information for qPCR Experiments (Bustin et al., 2009)
- 60+ checklist items (essential vs desirable)
- Primer design validation: T_m , GC%, amplicon length
- Amplification efficiency: 90-110% acceptable range
- Standard curve analysis with R-squared tracking

3. MIAME Standards

- Minimum Information About Microarray Experiment (Brazma et al., 2001)
- 27-item compliance checklist
- GEO submission: SOFT format generator
- ArrayExpress submission: IDF/SDRF format generator
- Sample and hybridization documentation

4. FAIR Principles

- Findable: DOI, ORCID, ROR persistent identifiers
- Accessible: HTTP/HTTPS protocols, authentication support
- Interoperable: Dublin Core, DataCite, Schema.org metadata
- Reusable: CC-BY-4.0 licensing, W3C PROV provenance
- Weighted FAIR assessment scoring (Gold/Silver/Bronze)

5. MINSEQE Standards

- High-throughput Sequencing Experiment standards
- 33-item checklist for WGBS/RRBS methylation sequencing
- SRA (Sequence Read Archive) submission generator
- ENA (European Nucleotide Archive) XML generator
- Bisulfite conversion efficiency documentation

6. GATHER Standards

- Gene list reporting guidelines
- HGNC symbol standardization
- Pathway analysis: GO, KEGG, Reactome, MSigDB
- Enrichment method documentation (hypergeometric, GSEA)
- 17-item compliance checklist

7. REMARK Standards

- Tumor Marker Prognostic Studies (McShane et al., 2005)
- Biomarker cutoff: median, tertile, optimal, ROC-based
- C-index (concordance) calculation
- Bootstrap validation with 95% CI
- Calibration analysis: slope, intercept, Hosmer-Lemeshow

8. STARD Standards

- Diagnostic Accuracy Studies (Bossuyt et al., 2015)
- Sensitivity, Specificity, PPV, NPV with Wilson CI
- Likelihood ratios (LR+, LR-), Diagnostic Odds Ratio
- ROC-AUC with bootstrap confidence intervals
- DeLong test for AUC comparison
- 30-item compliance checklist

PREVIOUSLY IMPLEMENTED STANDARDS

- 9. PRISMA-NMA:** Systematic review + Network Meta-Analysis, 32-item checklist, PubMed search, RoB 2/ROBINS-I, CINeMA framework
- 10. Statistical Validation:** Bonferroni, FDR, Holm, Storey q-value; Bootstrap CI (Percentile, BCa); ROC-AUC, PR-AUC; Cohen's d, Hedge's g
- 11. Reproducibility:** Global seed management (Python, NumPy, PyTorch, TensorFlow); SHA256 checksums; Pipeline manifests; STROBE-ME (38 items), TRIPOD checklists

VALIDATION RESULTS

Module	Status	Key Metrics
EWAS	PASS	Bonferroni: 5.77e-08, DMR detection OK
MIQE	PASS	Efficiency: 100.9%, R-squared: 1.0000
MIAME	PASS	SOFT/IDF/SDRF generation OK
FAIR	PASS	Dublin Core metadata, CC-BY-4.0
MINSEQE	PASS	SRA/ENA XML generation OK
GATHER	PASS	Pathway enrichment functional
REMARK	PASS	Bootstrap CI: [0.66, 0.84]
STARD	PASS	AUC: 0.867, DeLong test OK

MODULE FILES

- modules/ewas_standards.py - EWAS reporting, DMR, methQTL
- modules/miqe_standards.py - qPCR validation, primer design
- modules/miame_standards.py - Microarray, GEO/ArrayExpress
- modules/fair_principles.py - FAIR assessment, metadata
- modules/minseque_standards.py - Sequencing, SRA/ENA
- modules/gather_standards.py - Gene lists, pathways
- modules/remark_standards.py - Biomarker validation
- modules/stard_standards.py - Diagnostic accuracy, ROC